



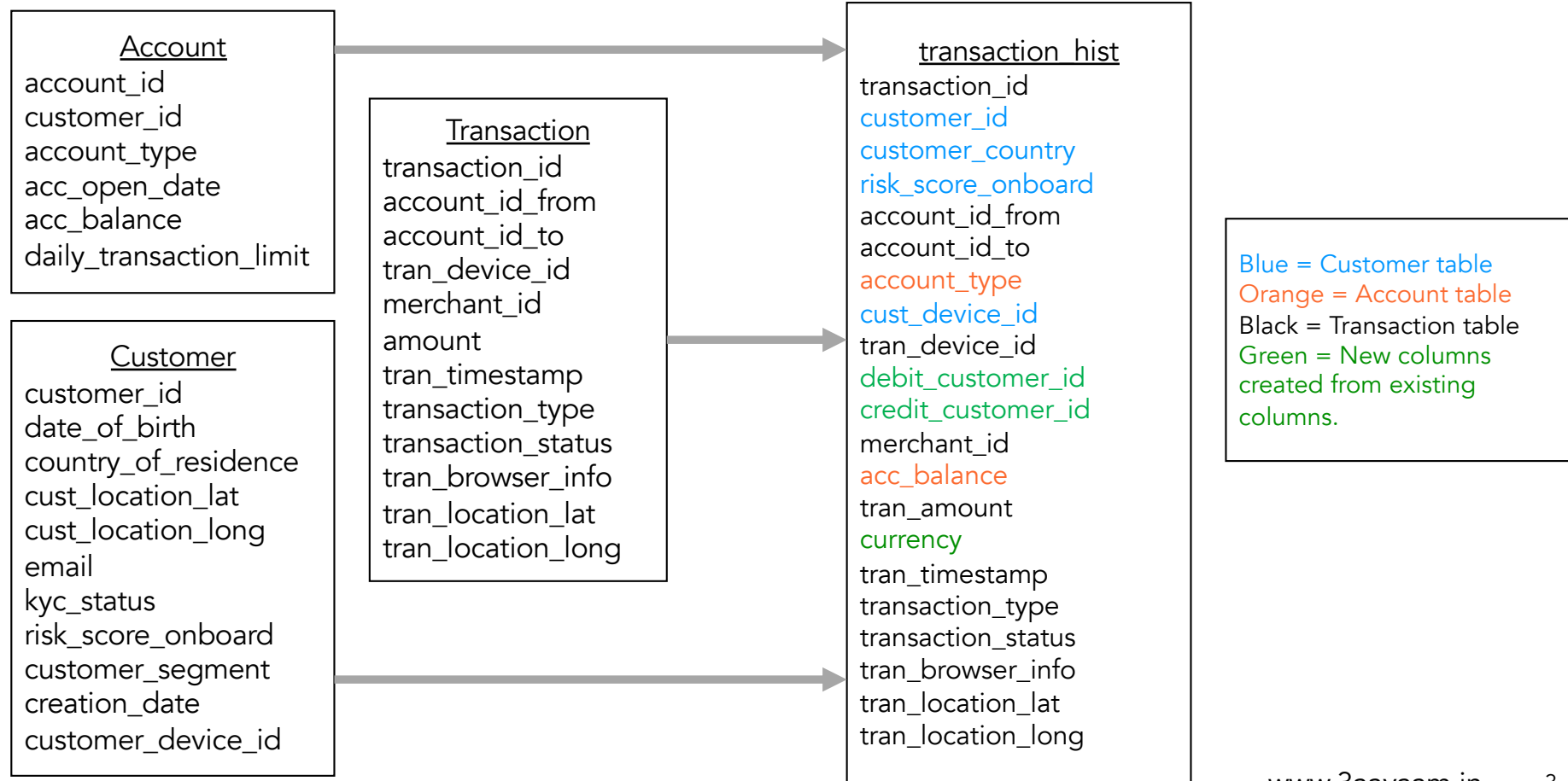
Project 10 Financial Fraud Detection System

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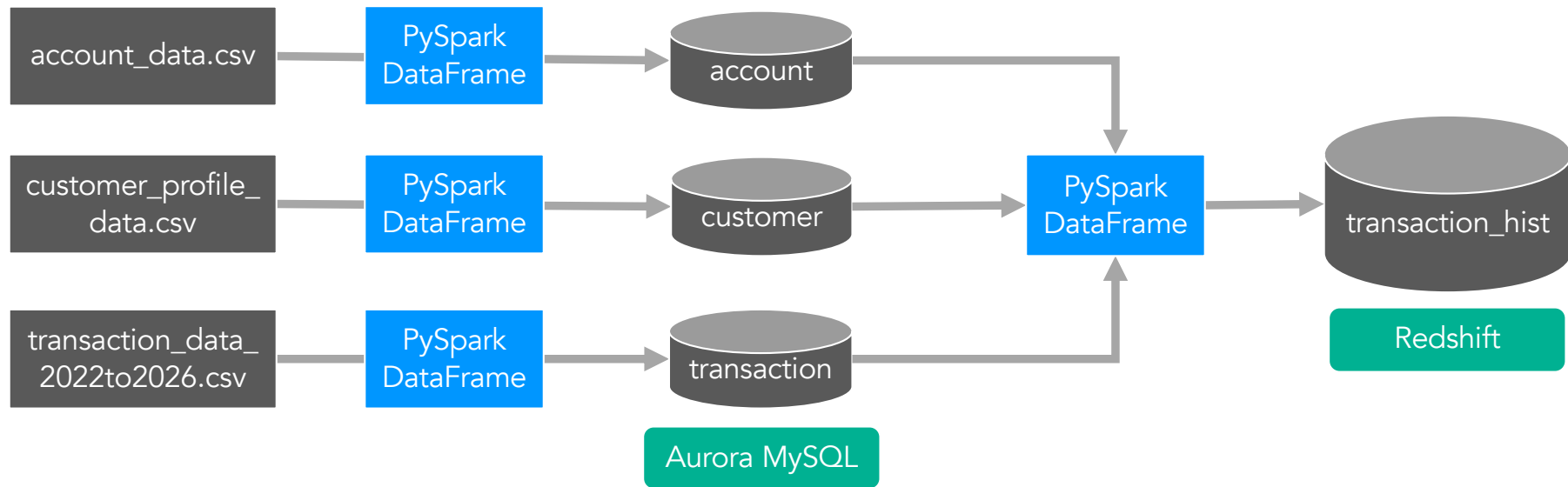
Fraud Detection System

- Financial Frauds - Card fraud, online payment fraud, money laundering, account takeover attempts etc.
- Entities involved:
 - Customer – Customer details with risk profiling. Dataset contains 100 customers.
 - Account – Account details held by customers.
 - Transaction – Historical transaction data.
- Considerations:
 - Each customer has only one device.
 - Each customer can have multiple accounts.
 - Some of the browsers are from dark web – Tor, Hyphanet, Torch, i2p, Haystak.
 - Current year is 2027.
 - Customer and Transaction LatLong has to be converted to distance in KM (Haversine formula).

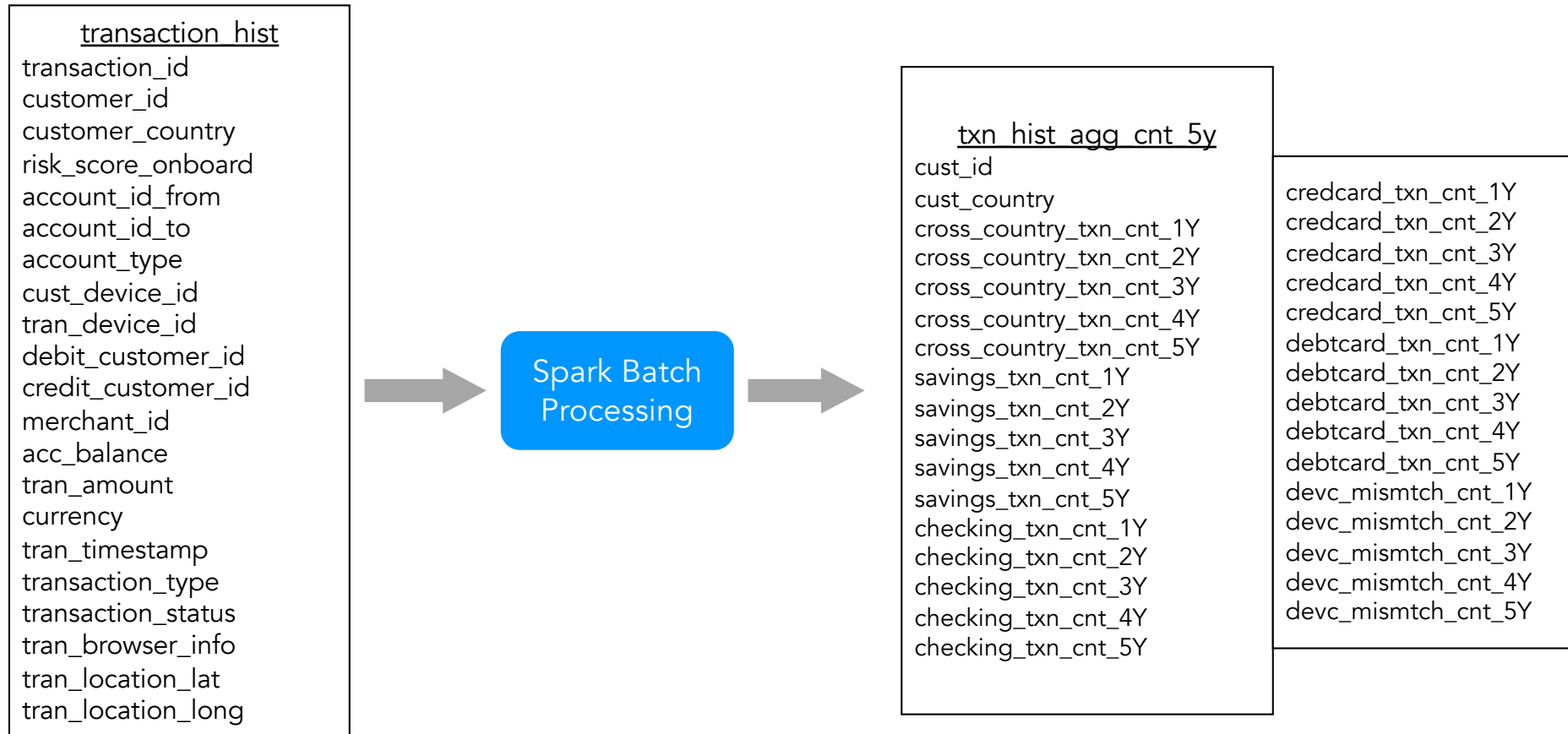
Transaction History Table



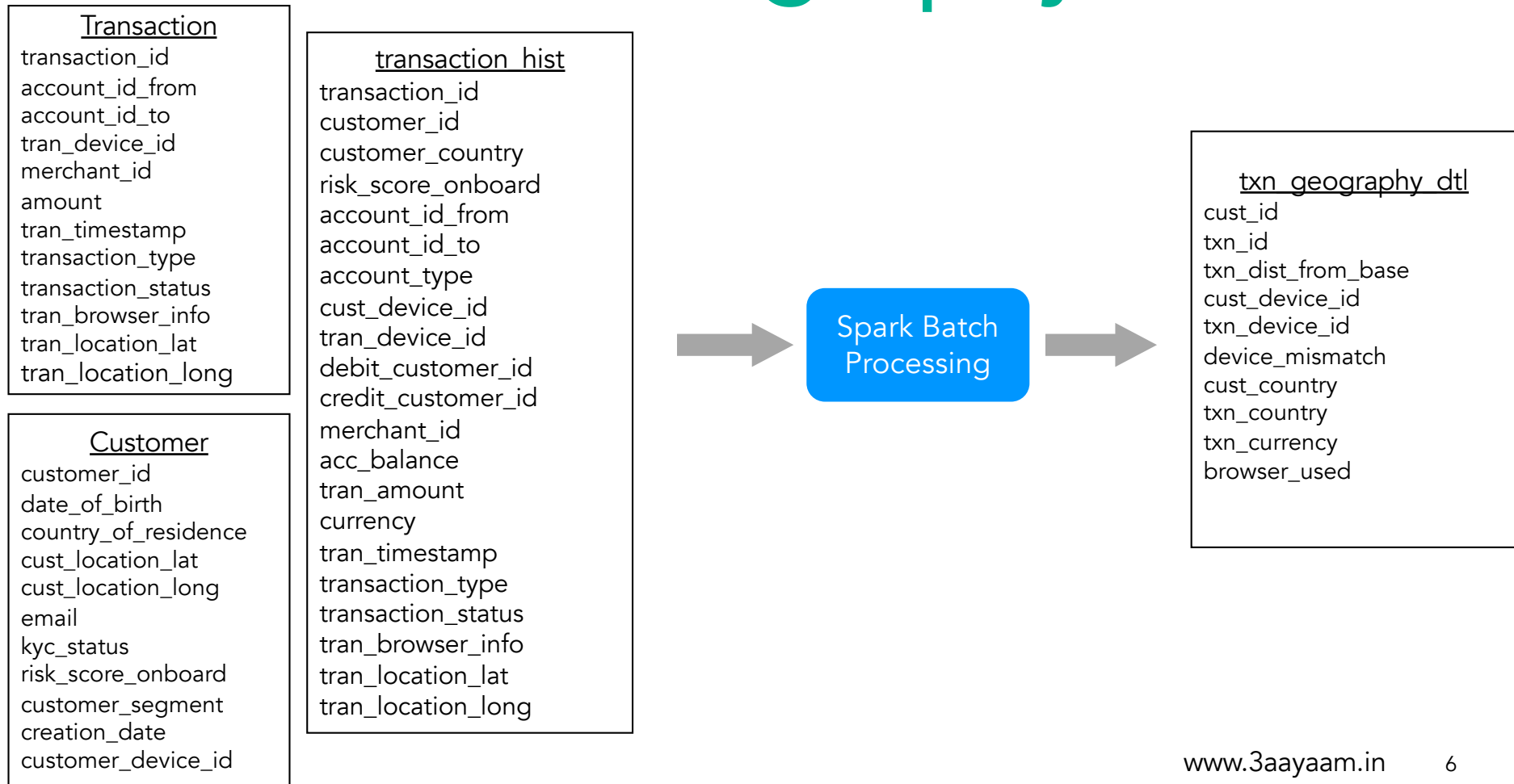
ETL 1



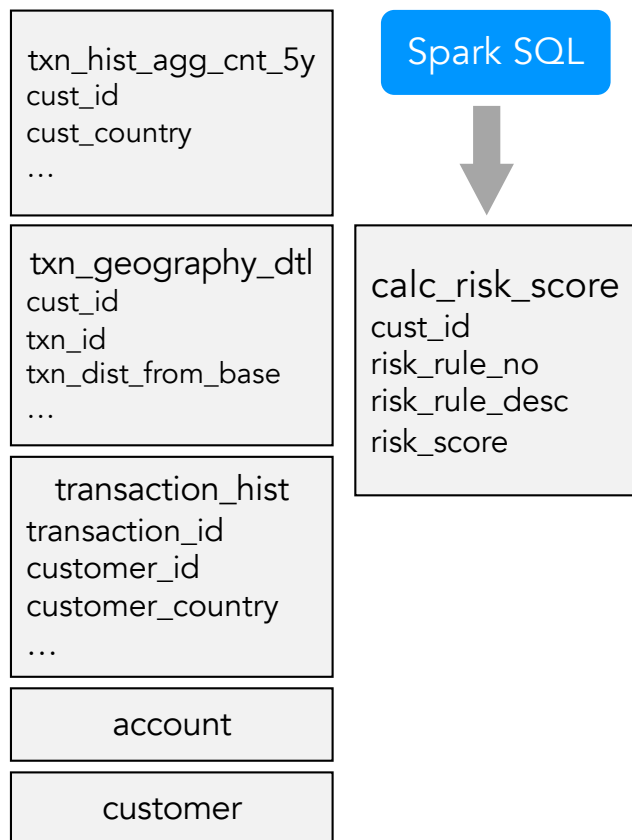
History Aggregation Count Table



Transaction Geography Table



Risk Score Table



Rule No.	Risk Rule	Risk Score
1	Cross country txn, from credit card with 5-year count more than 10, 15, 20	7, 8, 9
2	Cross country wire transfer txn where transferred amount is more than 25%, 35%, 40%, 45%, 50% of account balance	5, 6, 7, 8, 9
3	Same country, online txn with device 2 yr mismatch count more than 10, 15, 20, 25	4.5, 4.75, 5, 5.5
4	Same country, online txn and from credit card acc with amount more than 50%, 60%, 70% of account balance	6, 7, 8
5	Distance from base is more than 25 km using POS, ATM or P2P transaction with amount more than 15%, 20%, 25%, 30% of account bal	5, 6, 7, 8
6	"pending" KYC status and transaction amount is more than 5% of account balance	8
7	Cross country, any transaction with device mismatch, ATM & POS transaction type.	8
8	Cross country "reversed" transaction of "P2P" type	7
9	Credit card transaction count of 1Y, 2Y, 3Y is more than Debit card transaction.	6.5
10	Debit card transaction count of 1Y, 2Y, 3Y is more than the total of savings and checking transaction counts for the same year	6.5

Transaction Last Year (2026) Agg Table

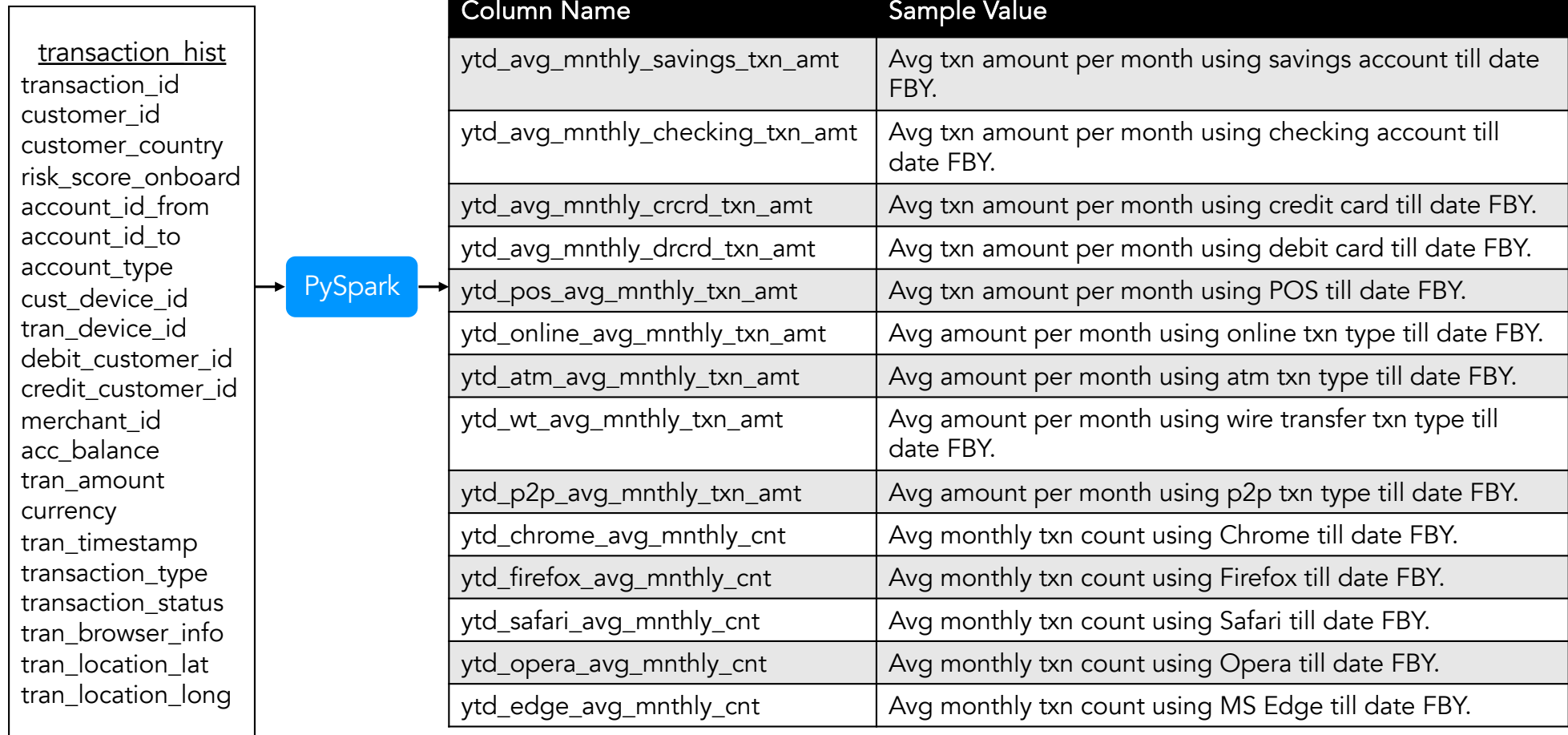
<u>transaction_hist</u>
transaction_id
customer_id
customer_country
risk_score_onboard
account_id_from
account_id_to
account_type
cust_device_id
tran_device_id
debit_customer_id
credit_customer_id
merchant_id
acc_balance
tran_amount
currency
tran_timestamp
transaction_type
transaction_status
tran_browser_info
tran_location_lat
tran_location_long

PySpark

txn_hist_agg_last_yr

Column Name	Sample Value
cust_id	Customer Id from batch file.
last_yr	The previous year (2025).
curr_yr	Current year (2026 - as per stream file).
mnth_no	01 to 12.
wk_no	01 to 05.
ytd_avg_mnthly_txn_amt	Avg txn amount per month till date from beginning of year (FBY).
ytd_max_mnthly_txn_amt	Max txn amount per month till date FBY.
ytd_min_mnthly_txn_amt	Min txn amount per month till date FBY.
ytd_avg_wkly_txn_amt	Avg txn amount per week till date FBY.
ytd_max_wkly_txn_amt	Max txn amount per week till date FBY.
ytd_min_wkly_txn_amt	Min txn amount per week till date FBY.
ytd_avg_yrly_txn_amt	Avg txn amount per year till date FBY.
ytd_max_yrly_txn_amt	Max txn amount per year till date FBY.
ytd_min_yrly_txn_amt	Min txn amount per year till date FBY.
ytd_total_txn_amt	Total txn amount till date FBY.

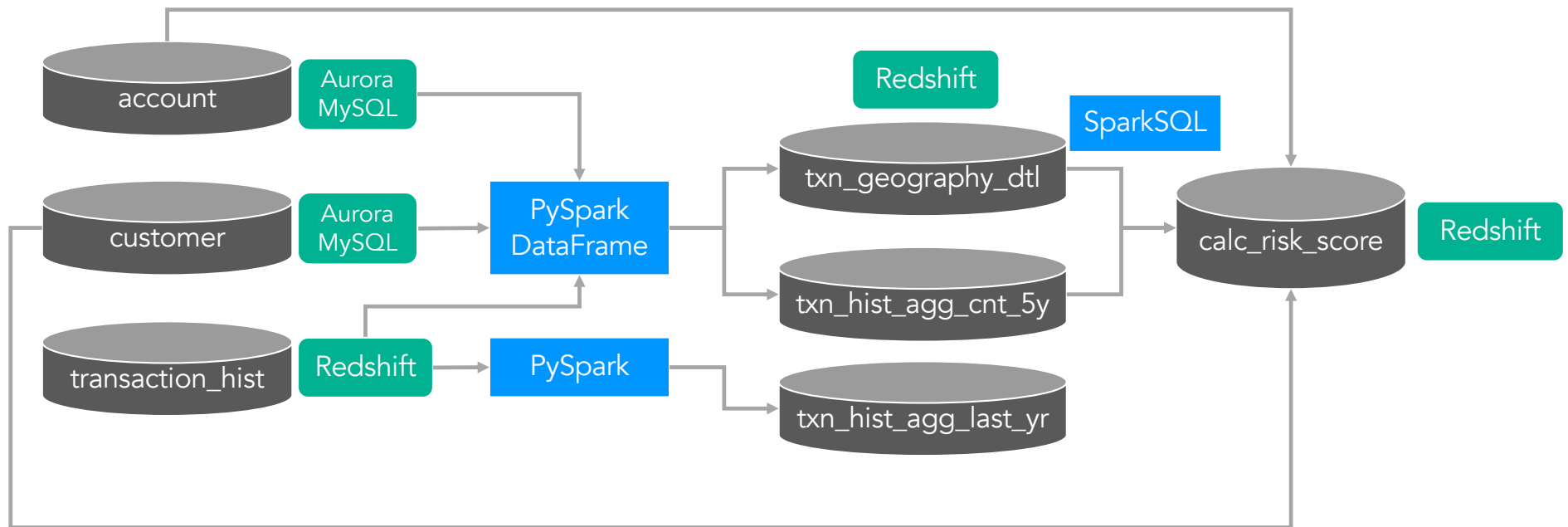
Transaction Last Year (2026) Agg Table



Column Description

Table Name	Column Name	Details
txn_hist_agg_cnt_5y	cross_country_txn_cnt_#Y	Count of transactions where source country and destination country were different. For last 1, 2, 3, 4, 5 yrs.
txn_hist_agg_cnt_5y	savings_txn_cnt_#Y	Count of transactions from savings account in the last 1, 2, 3, 4, 5 yrs.
txn_hist_agg_cnt_5y	checking_txn_cnt_#Y	Count of transactions from checking account in the last 1, 2, 3, 4, 5 yrs.
txn_hist_agg_cnt_5y	credcard_txn_cnt_#Y	Count of transactions from credit card in the last 1, 2, 3, 4, 5 yrs.
txn_hist_agg_cnt_5y	debtcard_txn_cnt_#Y	Count of transactions from debit card in the last 1, 2, 3, 4, 5 yrs.
txn_hist_agg_cnt_5y	devc_mismtch_cnt_#Y	No. of times where device id used in transaction is different than cust device in the last 1, 2, 3, 4, 5 yrs.
txn_geography_dtl	device_mismatch	Flag to

ETL 2

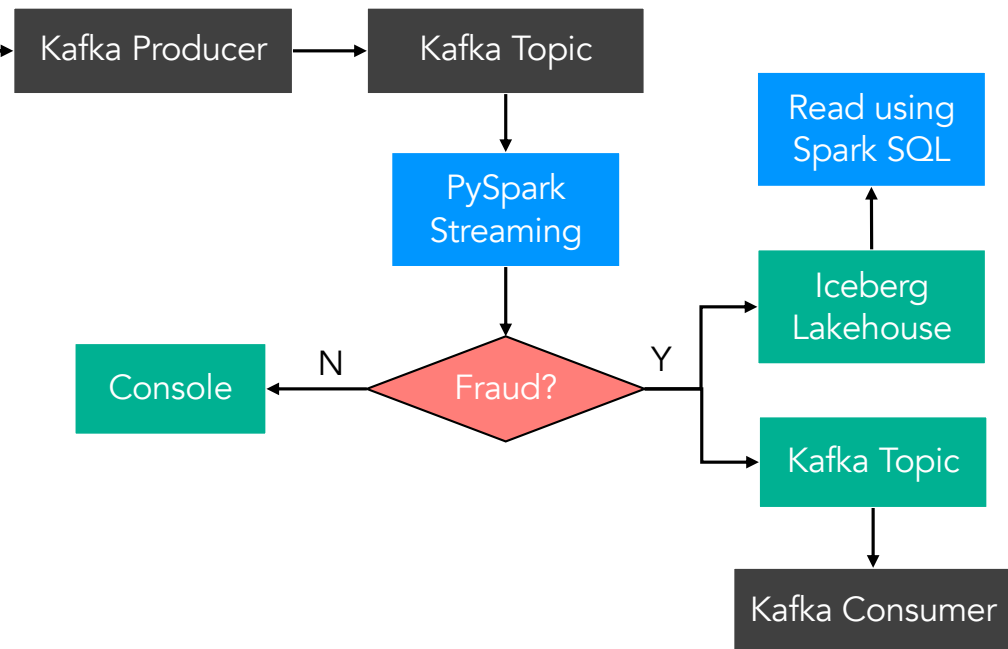


Real Time Fraud Detection

- Each transaction has to be verified for Fraud using rule-based filtering.
- Transactions will only be for 100 customers.

```
{  
  'transaction_id' : xxxxxx,  
  'account_id_from' : xxxxxx,  
  'account_id_to' : xxxxxx,  
  'tran_device_id' : xxxxxx,  
  'tran_amount' : xxxxxx,  
  'tran_timestamp' : xxxxxx,  
  'transaction_type' : xxxxxx,  
  'transaction_status' : xxxxxx,  
  'tran_browser_info' : xxxxxx,  
  'tran_location_lat' : xxxxxx,  
  'tran_location_long' : xxxxxx  
}
```

10 files with 10,000 records each



Fraud Detection Rule 1



Fraud?

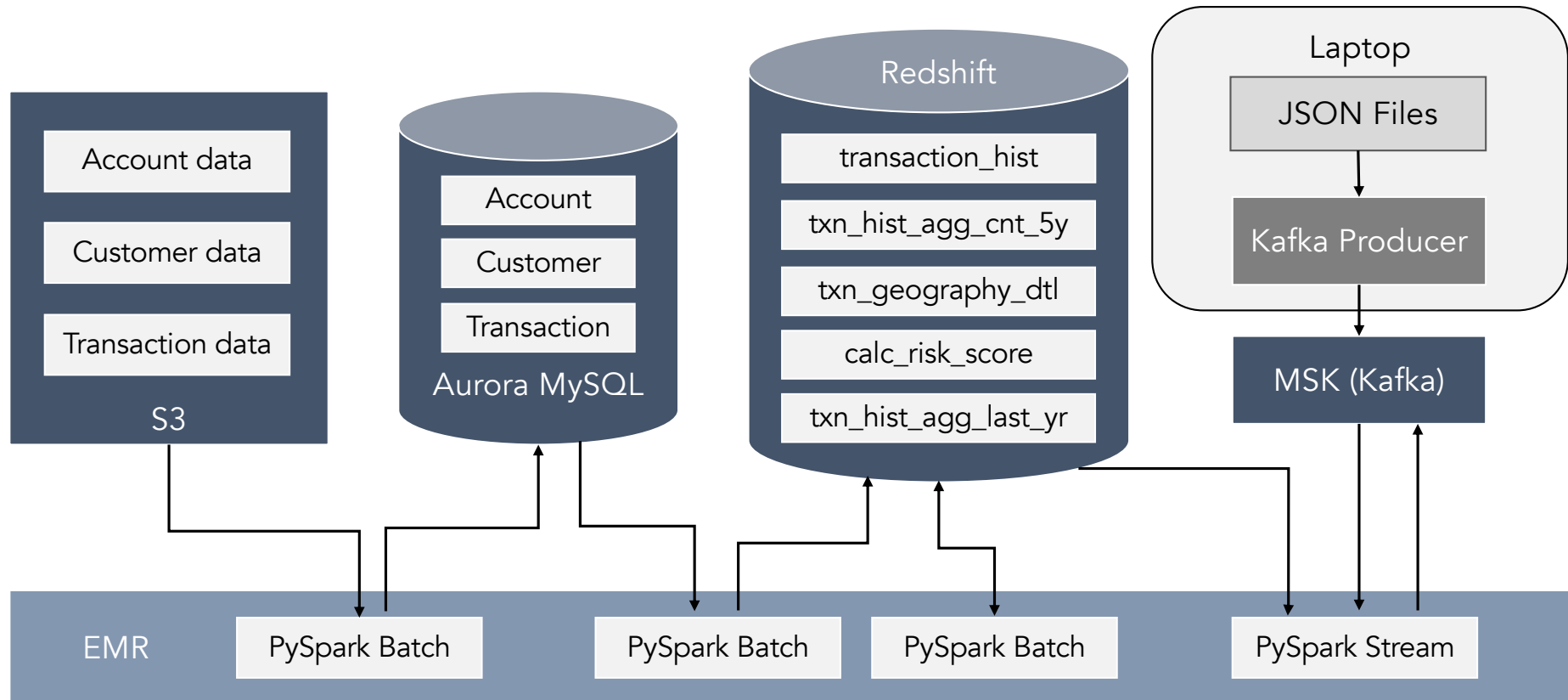
How to decide whether a transaction is fraud or not?

- Find risk score for transactions. Each transaction should be run against all the risk rules and find out the no. of rules it qualifies. And then find out the weighted risk score. Example below.
- Let's say txn 1 qualifies for Rule 1 with score 7, Rule 2 with score 5, Rule 5 with score 8, Rule 8 with score 7.
- Total risk score
 - Step 1: No. of qualified rules = 4 => Max Risk value = $4 \times 10 = 40$.
 - Step 2: Total risk value = Addition of all risk scores = $7 + 5 + 8 + 7 = 27$
 - Step 3: Final risk score for transaction = $(27/40) \times 10 = 6.75$
- If risk score:
 - Is more than 6 – Send to Kafka topic & write to Iceberg table with SMS Notification flag.
 - Is more than 7 – Send to Kafka topic & write to Iceberg table with CALL Notification flag.
 - Is more than 8 – Send to Kafka topic & write to Iceberg table with both CALL & SMS Notification flag.
 - Is more than 9 – Send to Kafka topic & write to Iceberg table with DECLINE Notification flag.

Fraud Detection Rule 2

- If Transaction amount :
 - Is more than YTD max yearly txn amount, DECLINE flag to Kafka and Iceberg.
 - Is within avg. and max. YTD yearly txn amount , but more than YTD max. monthly txn amount, CALL & SMS notification flag to Kafka & Iceberg.
 - Is within avg. and max. YTD yearly txn amount and within YTD avg. and max. monthly txn amount, SMS notification flag to Kafka & Iceberg.
 - Is more than 1.25 times of YTD avg. savings, checking and debit card amount, CALL notification flag to Kafka & Iceberg.
 - Is more than YTD avg. credit card amount, CALL notification flag to Kafka & Iceberg.
 - Is more than YTD avg. POS, Online, P2P amount, CALL notification flag to Kafka & Iceberg.
- Transaction should be allowed only when all combinations of Rule 1 & Rule 2 pass.

End-to-end Architecture (AWS)



Data Pipelines

- Think of the data pipeline for this.
- Load data from files to RDBMS tables.
- Extract, transform and load from RDBMS to DWH.
- Extract, transform and load from DWH to DWH.
- As there are tables that contain count, aggregation for the last 1 year to 5 years, these tables have to be loaded regularly to keep the data latest. Maybe once a month or quarter.
- Last year aggregation table has weekly aggregation columns, so this table should be refreshed at least once a week.
- Continuous streaming pipeline.